Hospital Patient Flow & Bed Capacity Planning

Team Briefing: What We've Built, Questions Solved, & Next Steps

Course: Applied Stochastic Models (ASM)	**GitHub:** github.com/Manas-Jain-24/applied-stochastic-models-hospital-flow
Data Grounding: 250,000 Real NYS SPARCS Discharges + MIMIC-IV Demo	**Code Status:** 33 Automated Unit Tests (100% Passing)

The 2-Minute Executive Summary

We are modeling hospital bed capacity, emergency overcrowding, and patient wait times using stochastic processes and queueing theory. Instead of waiting weeks for restricted data access, we calibrated our pipeline using **250,000 real hospital records** from New York State SPARCS and integrated the open-access **MIMIC-IV Demo** from PhysioNet. All 10 mathematical modules, discrete-event simulations, and 9 publication-grade figures are completely built, tested, and pushed to our GitHub repository.

The 11 Core Research Questions Solved

1. Does the textbook exponential assumption hold for hospital length of stay?

NO. Exponential service requires squared coefficient of variation cv^2 = 1.0. Our 250k real records revealed **cv^2 = 1.99 approx 2.0**. By the Allen-Cunneen M/G/c approximation, this recovery variance creates a **50% delay penalty** (Wq increases from 3.4 hrs to 5.1 hrs).

2. How many beds does a hospital actually need?

c* = 293 beds. For an offered load of a = 269.3 occupied bed-days, Erlang-B proves you cannot run at 100% capacity. Keeping blocking below 1% requires a safety buffer of at least **24 beds** to absorb stochastic arrival spikes.

3. How much does Emergency vs. Elective triage prioritization help?

It cuts emergency wait times by 8x without buying more beds. Two-class non-preemptive priority queueing ensures emergency patients wait only **1.5 hours** by safely shifting delay onto scheduled elective patients (12.3 hours).

4. Are hospital arrivals a steady Poisson process, or do they cluster?

Heavily clustered throughout the day. Index of Dispersion for Counts (IDC = 1.34 > 1.0) proved overdispersion. Arrivals peak at **2:12 PM** (3.25 arrivals/hr) and drop to 0.91 arrivals/hr at 2:12 AM. We modeled this with a diurnal Non-Homogeneous Poisson Process (NHPP) and Lewis-Shedler thinning.

5. What happens when Emergency Room holding space is capped?

In an M/M/c/K finite-buffer queue, expanding ED boarding holding beds from 0 to 20 cuts ambulance diversion from **12.4% down to 1.4%**, precisely quantifying the trade-off between ambulance access and patient boarding delay.

6. Where do patients go when they leave acute hospital care?

Using an **Absorbing Markov Chain** (fundamental matrix N = (I-Q)^-1), ICU patients spend an average of 1.48 ward visits before discharge, with a 10.1% in-hospital mortality risk and 33.2% transitioning to Skilled Nursing Facilities (SNF).

7. Can the hospital handle a winter flu / COVID surge?

NO, the system collapses. A +20% surge pushes traffic intensity to rho = 111% (> 1.0), triggering infinite queue growth. Hospitals must implement automated trigger rules to pause elective surgeries whenever occupancy reaches 95%.

8. How do hospital departments interact across an entire network?

We modeled an **Open Jackson Queueing Network** solving (I - P^T) Lambda = gamma. Patients transition between Emergency -> ICU -> Step-Down -> Med/Surg -> Discharge. Medical/Surgical is the primary bottleneck (rho = 92.2%), and total hospital dwell time averages **5.19 days** across all interconnected wards.

9. Can smart staffing shifts fix afternoon crowding without hiring more nurses?

YES, a 97.3% drop in peak delay with ZERO extra bed-hours! Pointwise Stationary Approximation (PSA) shows constant staffing causes afternoon wait times of 41.5 min. Shifting 25 beds from quiet night hours to peak afternoon hours drops wait time to **1.1 minutes**.

10. How should beds be partitioned between ICU, Emergency, and Surgery?

Proportional sizing fails; Fox's greedy marginal allocation saves 85.1% weighted blocking. Naive proportional sizing under-buffers the ICU (6.66% rejection rate). Optimal allocation provides +15 safety beds to ICU, reducing ICU rejection to **0.14%**.

11. What about MIMIC data?

We downloaded and parsed the official **MIMIC-IV Clinical Demo and MIMIC-IV-ED Demo** from PhysioNet (275 encounters, ICU length of stay, ESI triage levels 1-5). Our adapter (src/mimic_adapter.py) calibrated parameters into data/processed/mimic_empirical_parameters.json and is ready to process full MIMIC instantly.

How Teammates Can Clone and Run the Code (< 1 Second)

1. Clone the repository: git clone https://github.com/Manas-Jain-24/applied-stochastic-models-hospital-flow.git
2. Install packages: pip install -r requirements.txt
3. Run automated verification: python scripts/verify_all.py (validates all 33 unit tests and 9 figures in ~0.2s)
4. Open interactive notebook: jupyter notebook notebooks/01_stochastic_hospital_flow.ipynb

Recommended Division of Labor for the Final Submission

• **Teammate 1 (Math & Theory):** Review Sections 4 & 5 of the formal report (Erlang-B recursion, Erlang-C, Allen-Cunneen M/G/c, Jackson traffic equations, and Fox marginal allocation).

• **Teammate 2 (Literature Review):** Review Section 2 of the report (expanding citations on Linda Green, Armony, and de Bruin healthcare queueing papers).

• **You (Data & Presentation):** Present empirical results, the 9 publication figures, and assemble the 15-minute presentation slides matching Sir's exact guideline structure.